

EX6.

$$f: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

$$(x, y, z) \longmapsto (2x - y, -2x + y - 2z, x + y + 3z)$$

$f(e_1) \quad f(e_2) \quad f(e_3)$

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -2 & 1 & -2 \\ 1 & 1 & 3 \end{pmatrix} \begin{matrix} e_1 \\ e_2 \\ e_3 \end{matrix}$$

$$2) \det(A) = \begin{vmatrix} 2 & -1 & 0 \\ -2 & 1 & -2 \\ 1 & 1 & 3 \end{vmatrix}$$

$c_1 \leftarrow c_1 + 2c_2$

$$= \begin{vmatrix} 0 & -1 & 0 \\ 0 & 1 & -2 \\ 3 & 1 & 3 \end{vmatrix} = 1 \begin{vmatrix} 0 & -2 \\ 3 & 3 \end{vmatrix} = 1(0 - (-6)) = 6 \neq 0 \Rightarrow \text{invertible}$$

~~$$f(x, y, z) = A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$~~

$$3) \operatorname{com}(A) =$$

$$\begin{pmatrix} + \begin{vmatrix} 1 & -2 \\ 1 & 3 \end{vmatrix} - \begin{vmatrix} -2 & -2 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} -2 & 2 \\ 2 & 2 \end{vmatrix} \\ - \begin{vmatrix} -1 & 0 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ 1 & 3 \end{vmatrix} - \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} \\ + \begin{vmatrix} -1 & 0 \\ 1 & -2 \end{vmatrix} - \begin{vmatrix} 2 & 0 \\ -2 & -2 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ -2 & 1 \end{vmatrix} \end{pmatrix}$$
$$= \begin{pmatrix} 5 & 4 & -3 \\ 3 & 6 & -3 \\ 2 & 4 & 0 \end{pmatrix}$$

$$t_{\text{com}(A)} = \begin{pmatrix} 5 & 3 & 2 \\ 4 & 6 & 4 \\ -3 & -3 & 0 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \cdot t_{\text{com}(A)}$$

$$= \frac{1}{6} \begin{pmatrix} 5 & 3 & 2 \\ 4 & 6 & 4 \\ -3 & -3 & 0 \end{pmatrix}$$

$$A^{-1} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{5}{6}x + \frac{1}{2}y + \frac{1}{3}z \\ \frac{2}{3}x + y + \frac{2}{3}z \\ -\frac{1}{2}x - \frac{1}{2}y \end{pmatrix}$$

$$\Rightarrow f^{-1}(x, y, z) = \left(\frac{5}{6}x + \frac{1}{2}y + \frac{1}{3}z, \frac{2}{3}x + y + \frac{2}{3}z, -\frac{1}{2}x - \frac{1}{2}y \right)$$

16.

$$f: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

$$(x, y, z) \longmapsto (2x - y, -2x + y - 2z, x + y + 3z)$$

$f(e_1) \quad f(e_2) \quad f(e_3)$ $c_1 \quad c_2 \quad c_3$

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -2 & 1 & -2 \\ 1 & 1 & 3 \end{pmatrix} \begin{array}{l} e_1 \\ e_2 \\ e_3 \end{array}$$

4)

$A \in \mathcal{M}_3(\mathbb{R})$ inversible

ssi $\det(A) \neq 0$
inversible

$\Rightarrow$ inversible

$$\text{Com}(A)^{-1} = \frac{1}{\det(A)} {}^t A$$

$$= \frac{1}{6} \begin{pmatrix} 2 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 3 \end{pmatrix}$$

5)

$$\det_P(v_1, v_2, v_3) = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ -1 & -1 & 0 \end{vmatrix}$$

$c_2 \leftarrow c_2 - c_1, c_3 \leftarrow c_3 - c_1$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & -3 \\ -1 & -2 & -1 \end{vmatrix} = 1 \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix}$$

$$= -1 \neq 0 \dots$$

$\Rightarrow B^2 = (v_1, v_2, v_3)$ base de $\mathbb{R}^3$

6)

$$P_{B \rightarrow B^2} = \begin{pmatrix} v_1 & v_2 & v_3 \\ 1 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix} \begin{matrix} e_1 \\ e_2 \\ e_3 \end{matrix}$$

$$P_{B^2 \rightarrow B} = \begin{pmatrix} e_1 & e_2 & e_3 \\ 1 & 1 & 1 \\ -1 & -1 & -2 \\ 1 & 0 & 1 \end{pmatrix} \begin{matrix} v_1 \\ v_2 \\ v_3 \end{matrix}$$

$$\begin{cases} v_1 = e_1 + e_2 - e_3 & (1) \\ v_2 = e_1 + e_3 & (2) \\ v_3 = e_1 - e_2 & (3) \end{cases}$$

$$\Leftrightarrow \begin{cases} (1) - (2) \Rightarrow \boxed{e_2 = v_1 - v_2} \\ (3) \Rightarrow \boxed{e_1 = v_3 + e_2} = \boxed{v_3 + v_1 - v_2} \\ \quad \quad \quad = \boxed{v_1 - v_2 + v_3} \end{cases}$$

$$(2) \Rightarrow e_3 = e_1 - v_2 = v_3 + v_1 - v_2 - v_2$$

$$\boxed{e_3 = v_3 + v_1 - 2v_2}$$

$$e_3 = v_1 - 2v_2 + v_3$$

X6.

$$4) D = \text{mat}(f, B')$$

$$= \begin{pmatrix} f(v_1) & f(v_2) & f(v_3) \\ 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{matrix} v_1 \\ v_2 \\ v_3 \end{matrix}$$

$$f(v_1) = f(1, 1, -1) = (1, 1, -1) = v_1$$

$$f(v_2) = f(1, 0, -1) = (2, 0, -2) = 2v_2$$

$$f(v_3) = 3v_3$$

$$A = \text{mat}(f, B)$$

$$D = \text{mat}(f, B')$$

$$P = B \rightarrow B'$$

$$D = P^{-1} A P$$

$$A = P \cdot D \cdot P^{-1}$$

$$9) D^m = \begin{pmatrix} 1^m & & 0 \\ & 2^m & \\ 0 & & 3^m \end{pmatrix}$$

car $D = \text{diag}(1, 2, 3)$

10) pour $m=0$

$$A^0 = P \cdot D^0 \cdot P^{-1} \text{ Vrai}$$

pour $m \in \mathbb{N}$

On suppose que $A^m = P \cdot D^m \cdot P^{-1}$
et montrons que $A^{m+1} = P \cdot D^{m+1} \cdot P^{-1}$

$$A^{m+1} = \overset{m}{A} \cdot A$$

$$= P \cdot D^m \cdot P^{-1} \cdot P \cdot D \cdot P^{-1}$$

$$= P \cdot D^{m+1} \cdot P^{-1}$$

$\mathbb{C}$ $m \in \mathbb{N}$ V_1

$$A^m = P \cdot D^m \cdot P^{-1}$$

$$= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2^m & 0 \\ 0 & 0 & 3^m \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & -2 \\ 1 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2^m & 3^m \\ 1 & 0 & -3^m \\ -1 & -2^m & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & -2 \\ 1 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1-2^m+3^m & 1-2^m & 1-2^{m+1}+3^m \\ 1-3^m & 1 & 1-3^m \\ -1+2^m & -1+2^m & -1+2^{m+1} \end{pmatrix}$$

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \\ z_{n+1} \end{pmatrix} = A \begin{pmatrix} x_n \\ y_n \\ z_n \end{pmatrix}$$

$$U_{n+1} = A U_n$$

$$U_n = A U_{n-1}$$

$$U_{n-1} = A U_{n-2}$$

$$\vdots$$

$$U_1 = A U_0$$

$\Rightarrow$ On multiplie membre a membre
 $\Rightarrow U_n = A^n U_0, \forall n$

$$x_0 = z_0 = 1, y_0 = -1$$

$$\begin{cases} x_{n+1} = 2x_n - y_n \\ y_{n+1} = -2x_n + y_n - 2z_n \\ z_{n+1} = x_n + y_n + 3z_n \end{cases}$$

$$U_n = A^n \cdot U_0$$

$$= \begin{pmatrix} 1-2^n+3^n & 1-2^n & 1-2^{n+1}+3^n \\ 1-3^n & 1 & 1-3^n \\ -1+2^n & -1+2^n & -1+2^{n+1} \end{pmatrix}$$

$$= \begin{pmatrix} 1+2 \times 3^n - 2^{n+1} \\ 1-2 \times 3^n \\ -1+2^{n+1} \end{pmatrix}$$

$$\Rightarrow \begin{cases} x_n = 1 + 2 \times 3^n - 2^{n+1} \\ y_n = 1 - 2 \times 3^n \\ z_n = 2^{n+1} - 1 \end{cases}$$

$$y_n = 1 - 2 \times 3^n$$

$$z_n = 2^{n+1} - 1$$

$$\begin{pmatrix} 1-2^n+3^n & 1-2^n & 1-2^{n+1}+3^n \\ 1-3^n & 1 & 1-3^n \\ -1+2^n & -1+2^n & -1+2^{n+1} \end{pmatrix}$$